# Machine Learning

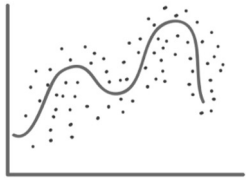
## Lecture 14: Clustering

Spring 2025

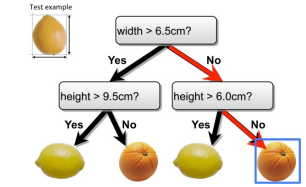
Instructor: Xiaodong Gu

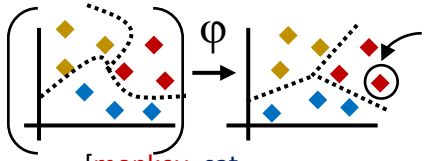


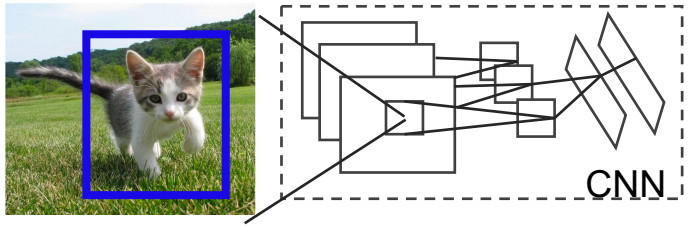
# What we have learned so far?



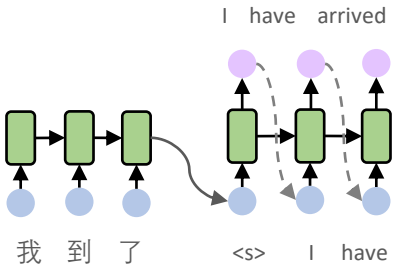
Regression




Classification




Generation

All previous lectures involve data labels (supervised learning)

# What if we have no labels?






| VISITORS | CONTACTS | LEADS | MQLS | BI_USER | CUSTOMERS | EXPECTED_ARR |
|---|---|---|---|---|---|---|
| 14K | 2.0K | 1.650K | 76.0 | 78.0 | 2 | 2.6M |
| ↑ 37.2% | ↑ 19.1% | ↑ 21.0% | ↑ 35.7% | ↑ 69.6% | ↑ N/A | ↑ 2.3% |

Medium | Source | Group Channel

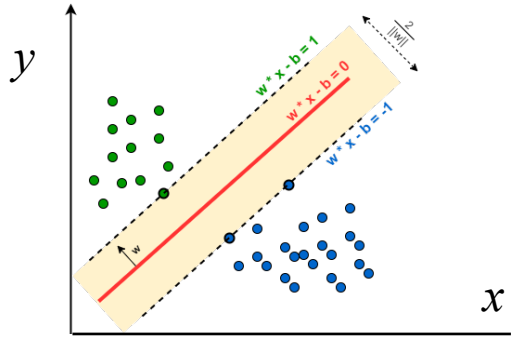
| GROUP CHANNEL | SOURCE | VISITORS | % Δ | LEADS | % Δ | LEADS TO MQLS CR | % Δ | MQLs | % Δ |
|---|---|---|---|---|---|---|---|---|---|
| Other | webinar | 5.996 | 31.1% ↑ | 152 | 76.7% ↑ | 8% | 69.7% ↑ | 12 | 200.0% ↑ |
| Organic Traffic | yandex | 3.424 | 18.8% ↑ | 102 | 18.6% ↑ | 2% | -15.7% ↓ | 2 | 0.0% |
| Referral Traffic | leadmagnet | 918 | 9.08% ↑ | 1.172 | 18.1% ↑ | 4% | 0.0% ↑ | 52 | 18.2% ↑ |
| Brand Traffic | leadmagnet | 722 | 65.6% ↑ | 2 | 58.3% ↑ | 5% | 0.0% ↑ | 2 | 0.0% |
| Offline | Event | 346 | -3.4% ↓ | 2 | -66.7% ↓ | 0% | 0.0% ↑ | 0 | 0.0% |
| Referral Traffic | evernote.com | 202 | 260.7% ↑ | 6 | -88.9% ↓ | 0% | 0.0% ↑ | 0 | 0.0% |



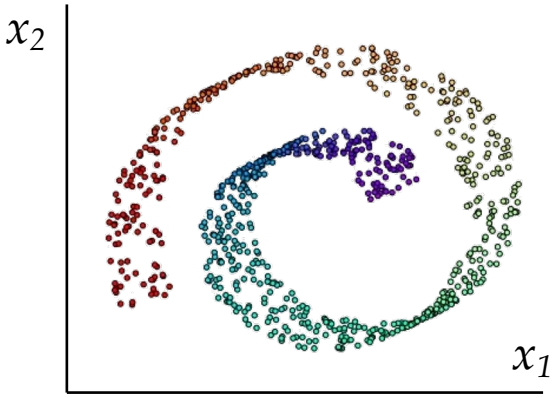

Can we still learn something from the data?

# Supervised vs Unsupervised Learning

- Machine Learning
  - Supervised Learning
  - **Unsupervised Learning**
    - Clustering
    - Dimensional Reduction
    - Density Estimation
    - …
  - Reinforcement Learning




Supervised: (have labels)




Unsupervised: (No labels)

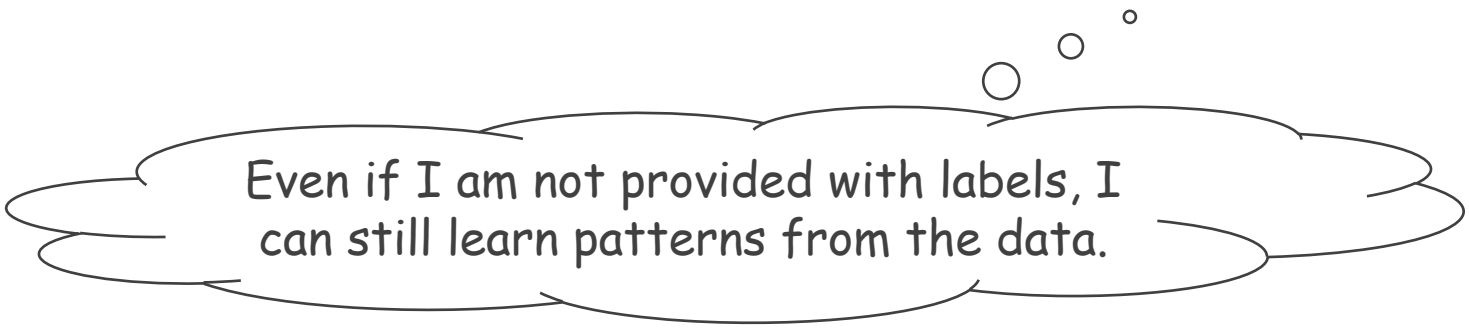
# Supervised vs Unsupervised Learning

## Supervised Learning

discover patterns in the data that relate data attributes **with a target (class)** attribute.

## Unsupervised Learning

The data have **no target attribute**. We want to explore the data to find some **intrinsic structures**.



# Unsupervised Learning



## Clustering

Learning similarity

- K-means (✓)
- Gaussion Mixture



## Generative

Density estimation

- GAN
- VAE
- Diffusion



## Dimensional Reduction

Data compression

- Principal component analysis
- Auto-encoder

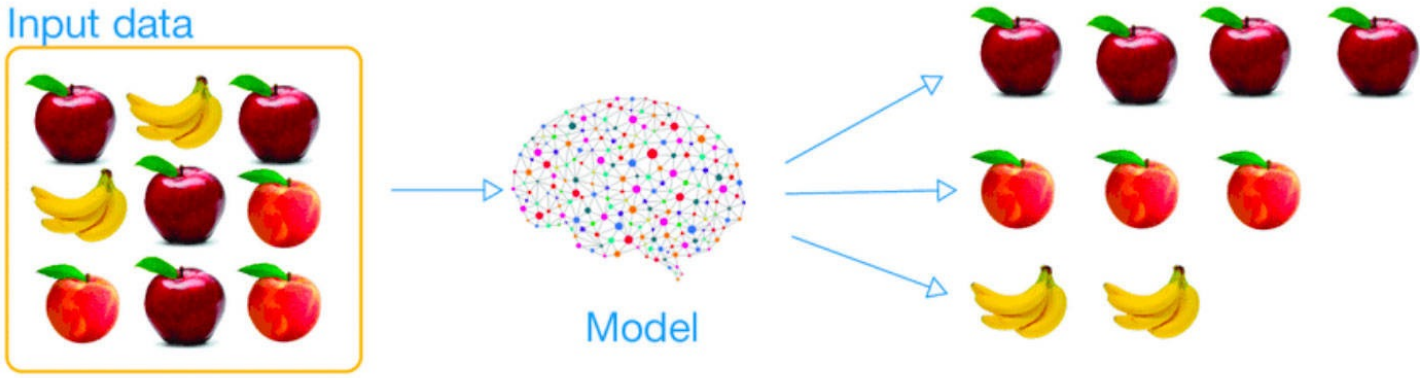
# Today

- Clustering
- K-means



# What is Clustering?

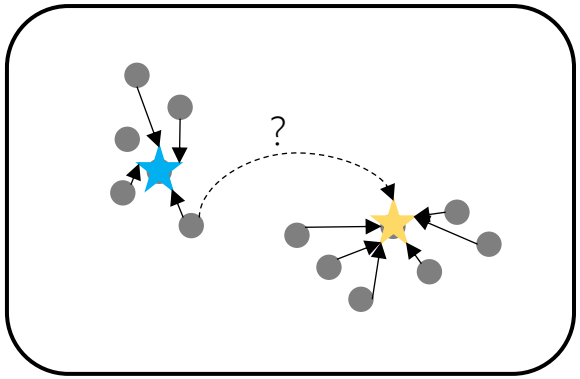
- **Clustering** = the organization of unlabeled data into similarity groups (called clusters).
- A cluster (簇) is a collection of data items which are "**similar**" between them, and "**dissimilar**" to data items in other clusters.




Q: How can we characterize "Similar" and "Dissimilar"?

# How to Represent Clusters ?

- Suppose we have figured out the clusters for a dataset (e.g., by eyes), how can we represent them?




– For each cluster, elect a "**center**" point which best represents the whole cluster.

– Each data instance is assigned with a "label" indicating its **membership** to each cluster.

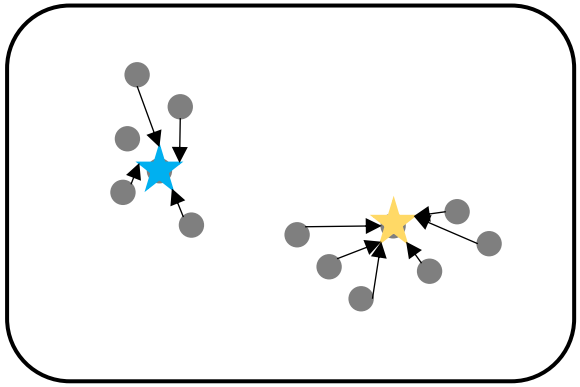


"Similar" or "Dissimilar"?

- The centroid must be more **similar** to instances in the represented cluster than other instances.
- Instances within a cluster are more **similar** to the centroid of the same cluster than centroids of other clusters.

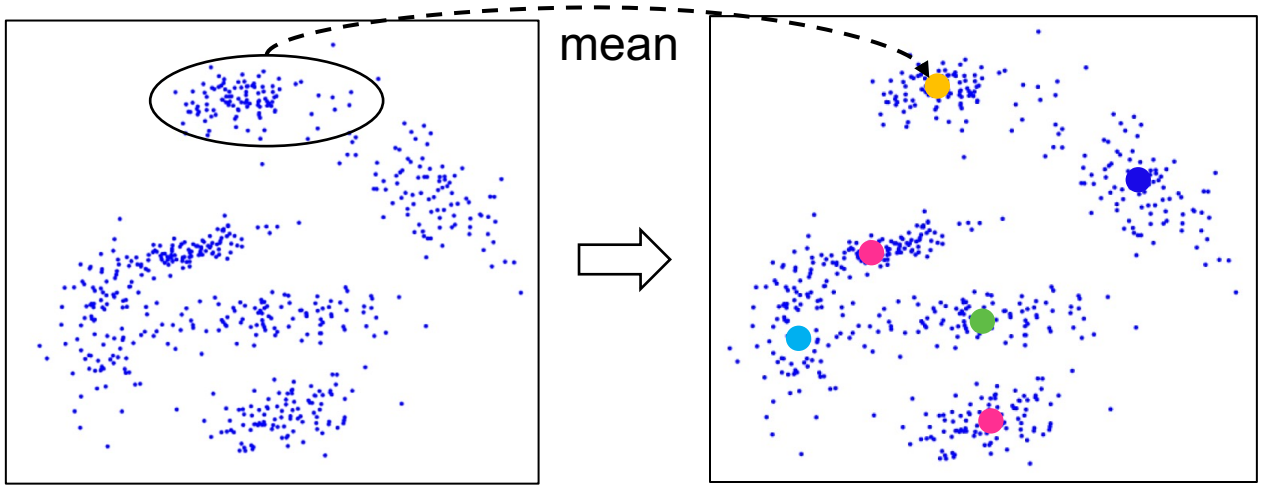
# A Machine Learning Perspective

- What can be optimized in this "center-membership" framework?
  - The "**centers**" should be voted according to the pairwise distances of the instances in the same cluster. [**center update**]
  - Having obtained the centroids, we need to optimize the "**membership**" (label) of each data point. [**label update**]
  - The above two steps can be repeated alternatively.



# K-Means [MacQueen, 1967]

- Partition the data into ***k*** clusters, where the **center** for each cluster is defined as the **mean** of all instances within the cluster, called **centroid**.



# Problem Formulation




- Given a dataset $D = \{x_1,\ldots,x_N\}$, where $x_i \in \mathbb{R}^d$ is a $d$ dimensional vector.
- Find K reference vectors which best represent the clusters
- Assign each $x^{(i)}$ with a variable $m_i \in \{1,\ldots,K\}$ to indicate its cluster membership.

# Steps

1. Choose $k$ (random) data points (seeds) to be the initial centroids, cluster centers.
2. Assign each data point to the closest centroid.

$$L_2(x, \mu^k) = \|x - \mu^k\| = \sqrt{\sum_{m=1}^{d} (x_i - \mu_m^k)^2}$$

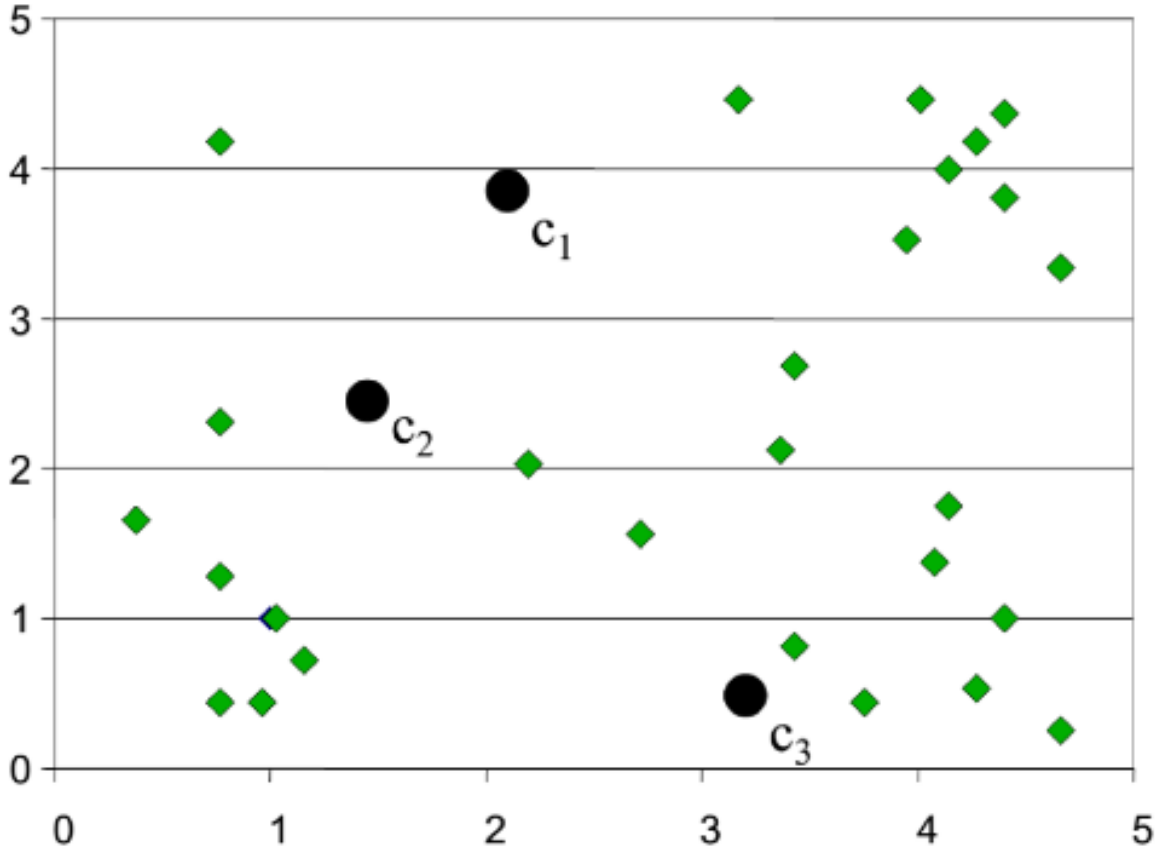
3. Re-compute the centroids using the current cluster memberships.

$$\mu^k = \frac{1}{C_k} \sum_{x \in C_k} x$$

4. Repeat steps 2 and 3 until a convergence criterion is met.

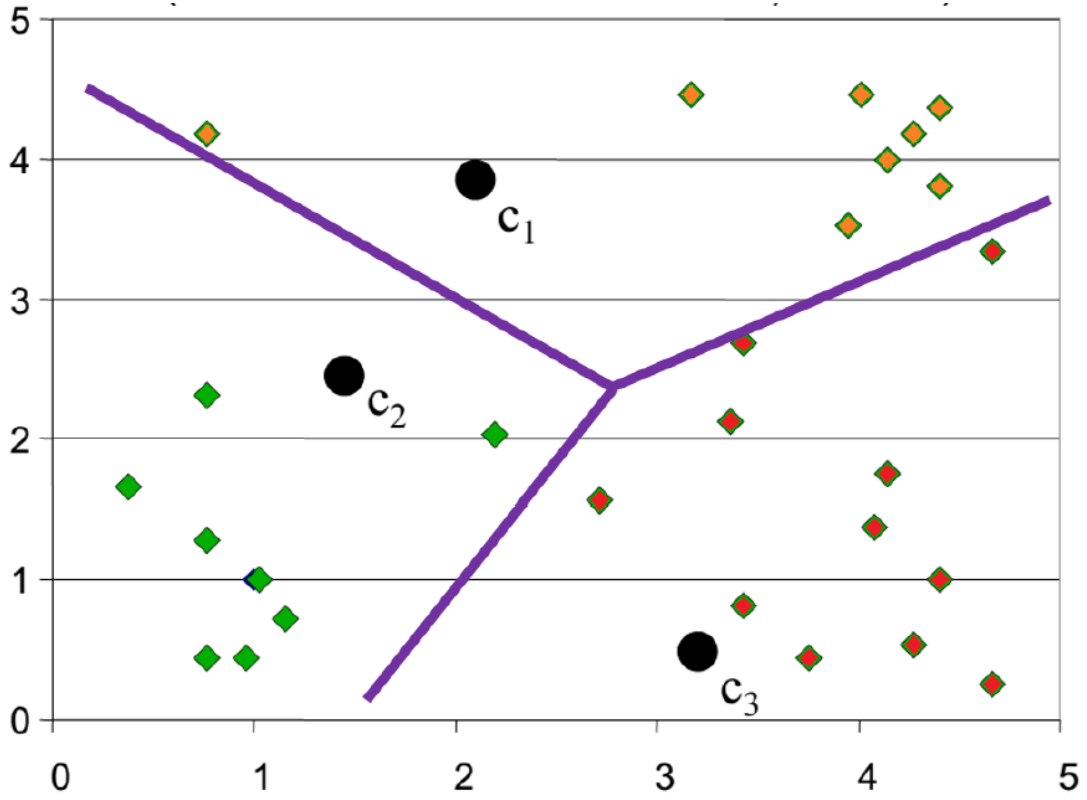
# Example – Step1

**Randomly initialize the cluster centers.**



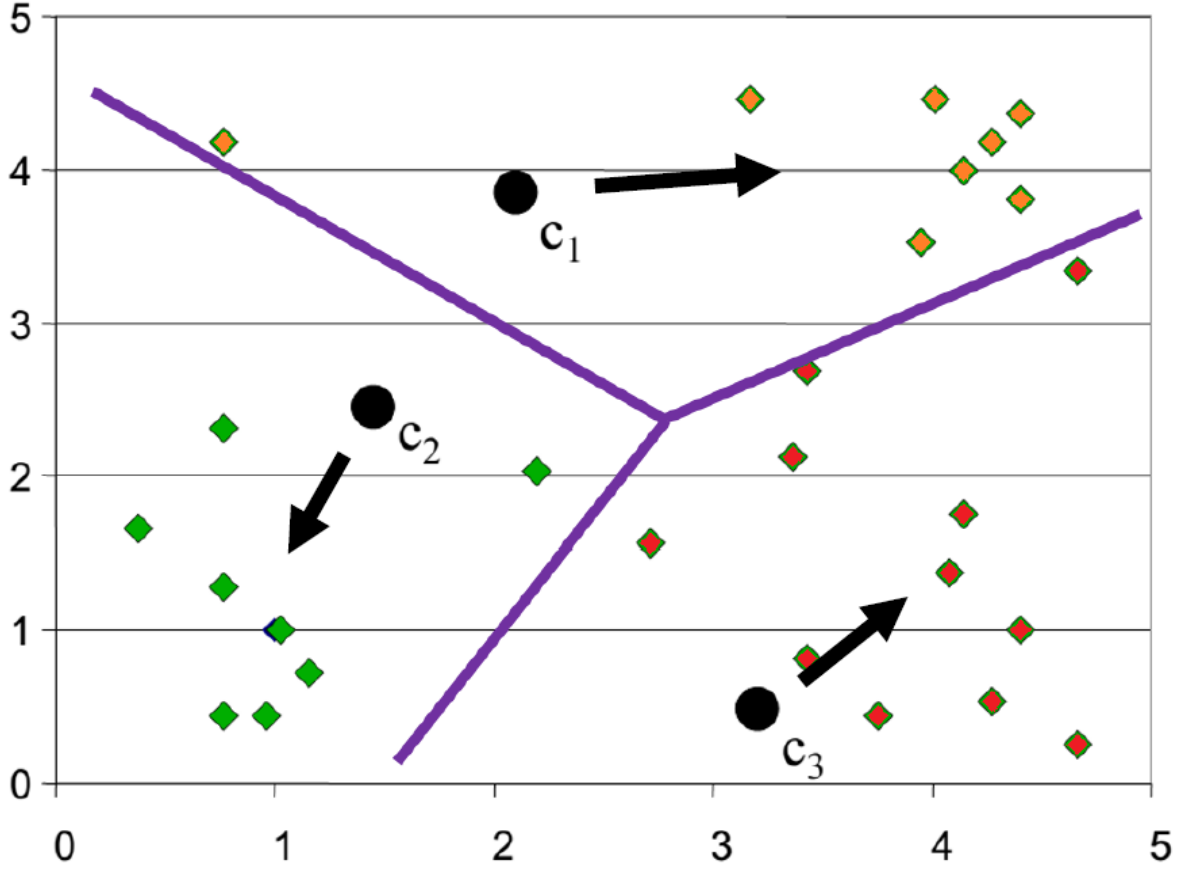
# Example – Step2

**Determine cluster membership for each input**
("winner-takes-all" inhibitory circuit)



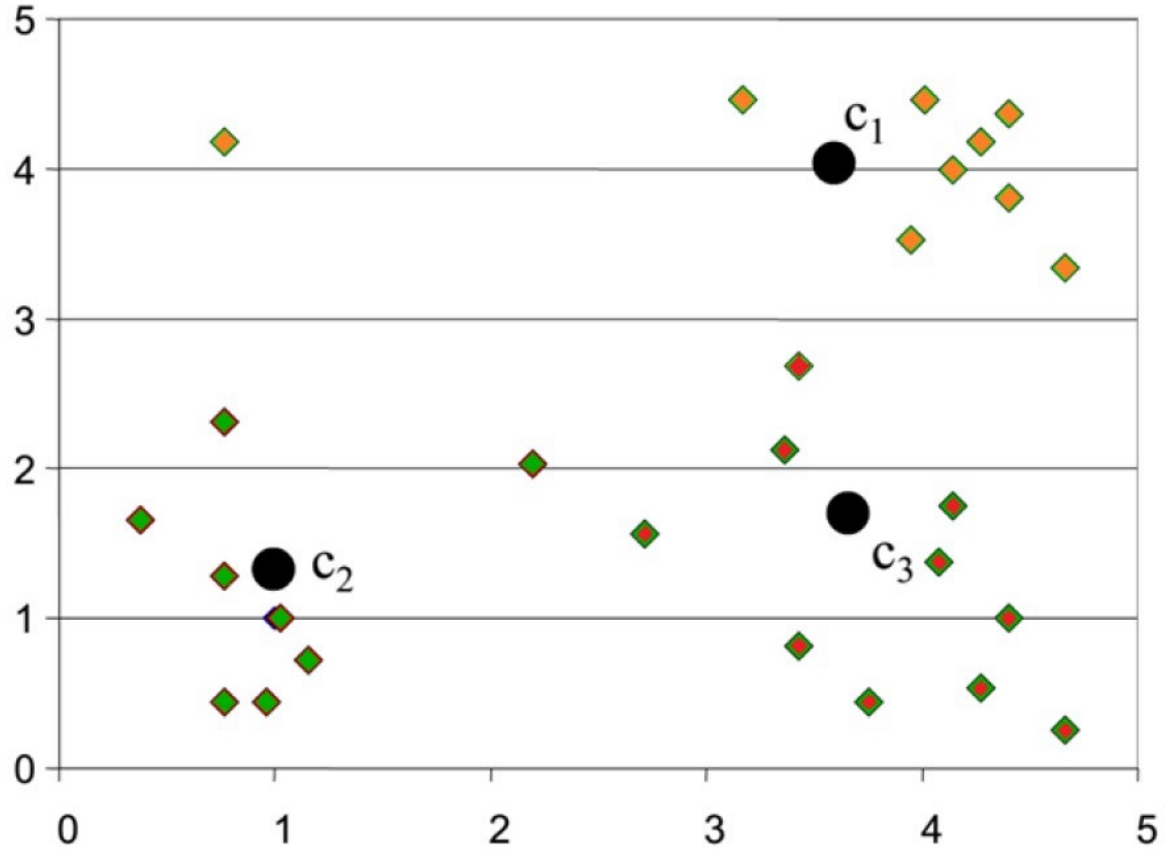
# Example – Step3

**Re-estimate cluster centers** (adapt synaptic weights)



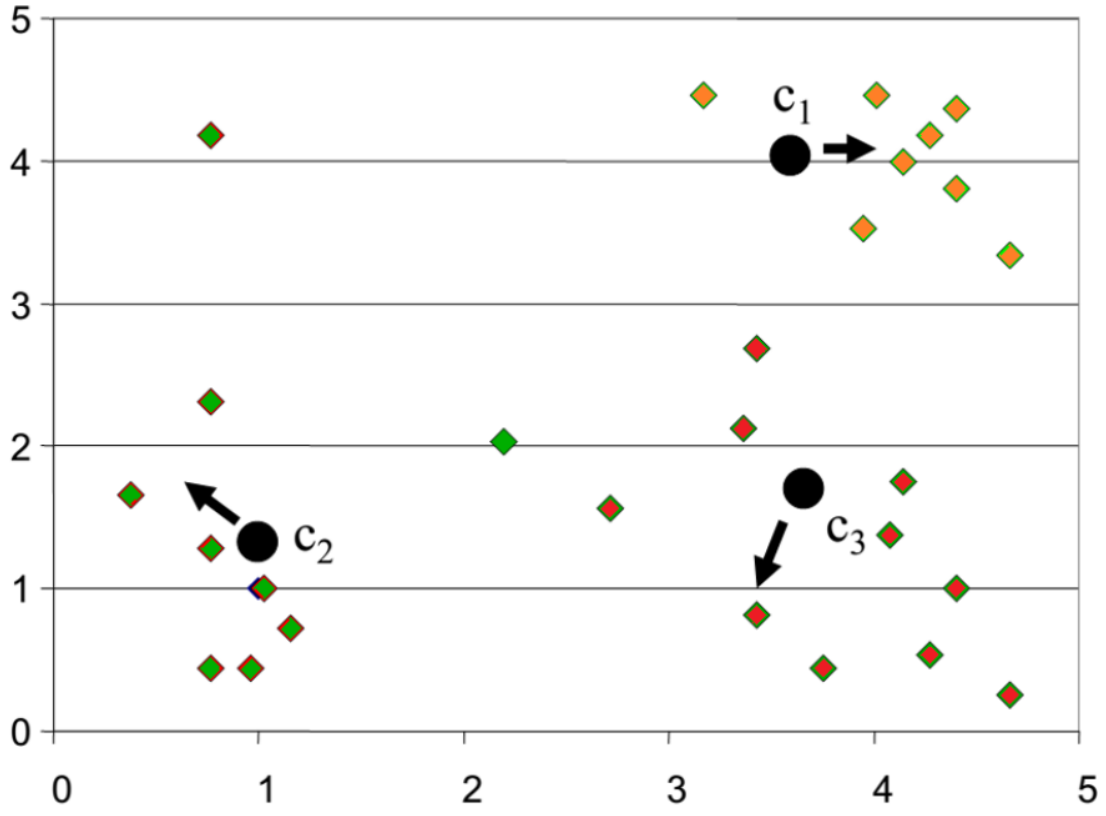
# Example

Result of the first iteration



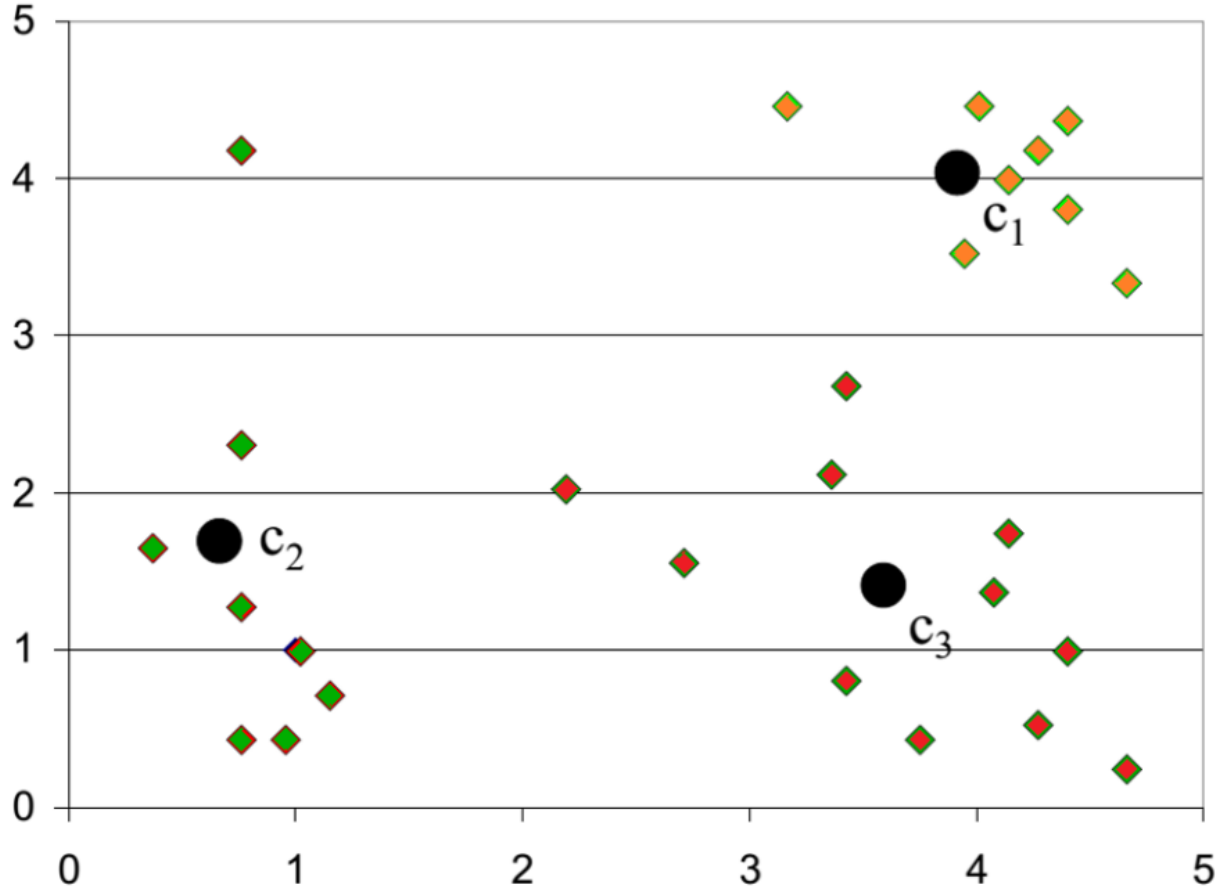
# Example

The second iteration



# Example

Result of the second iteration

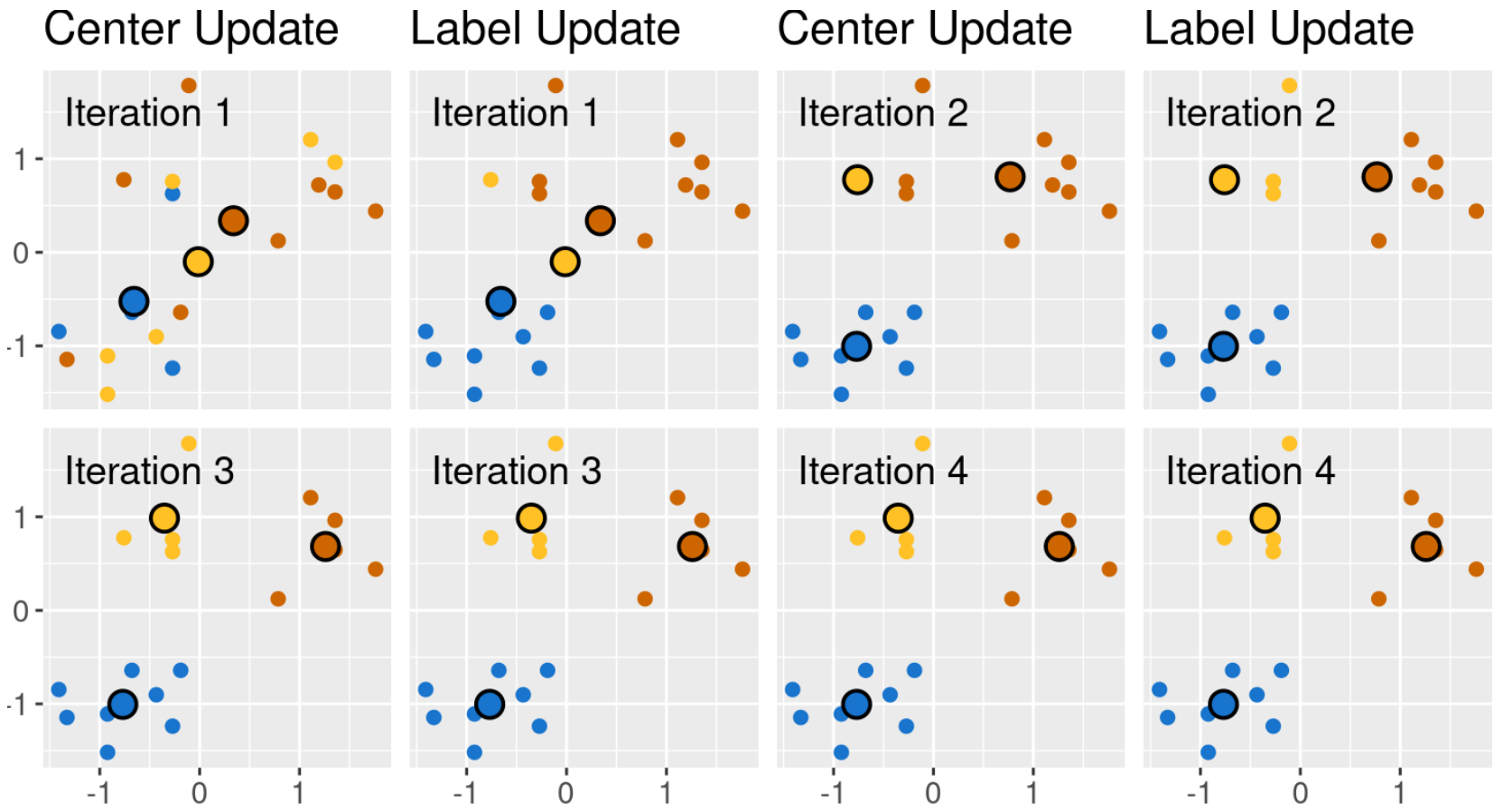


# Convergence Criterion

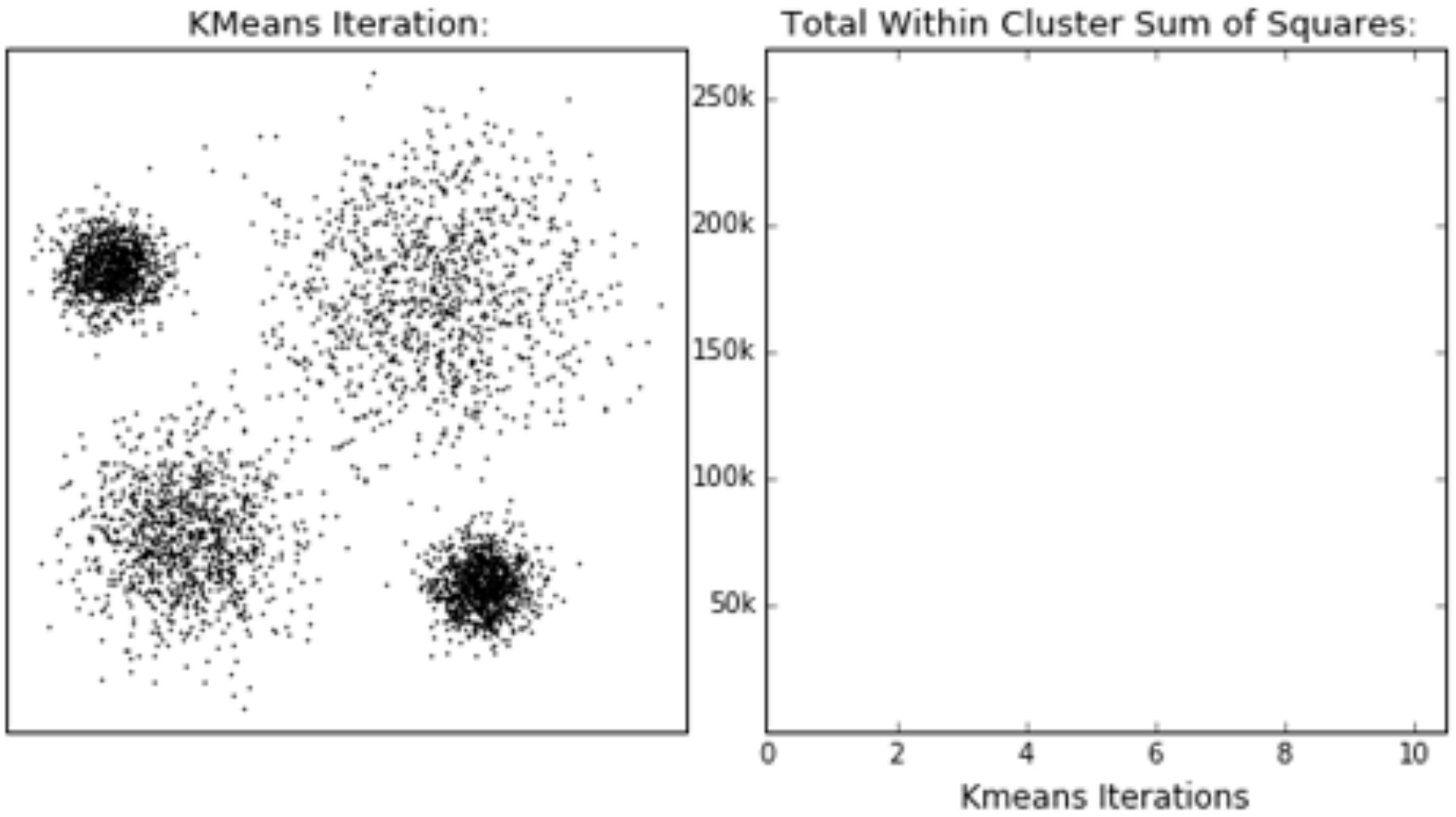
- No (or minimum) re-assignments of data points to different clusters, *or*
- No (or minimum) change of centroids, or
- Minimum decrease in the **sum of squared error** (SSE):

$$\min_{\{\mu^k\}_{k=1}^K} \sum_{k=1}^{K} \sum_{x \in C_k} L(x - \mu^k) \qquad \mu^k = \frac{1}{C_k} \sum_{x \in C_k} x$$

# Example – A Big Picture



# Example: Live K-Means



# Game Time







Which food should I taste first for each cuisine?

# Time Complexity

- Assume computing distance between two instances is $O(d)$ where $d$ is the dimensionality of the vectors.
- **Reassigning clusters**: $O(knd)$ distance computations.
- **Computing centroids**: Each instance vector gets added once to some centroid: $O(nd)$.
- Assume these two steps are each done once for $I$ iterations: $O(Iknd)$

# Local Optimum

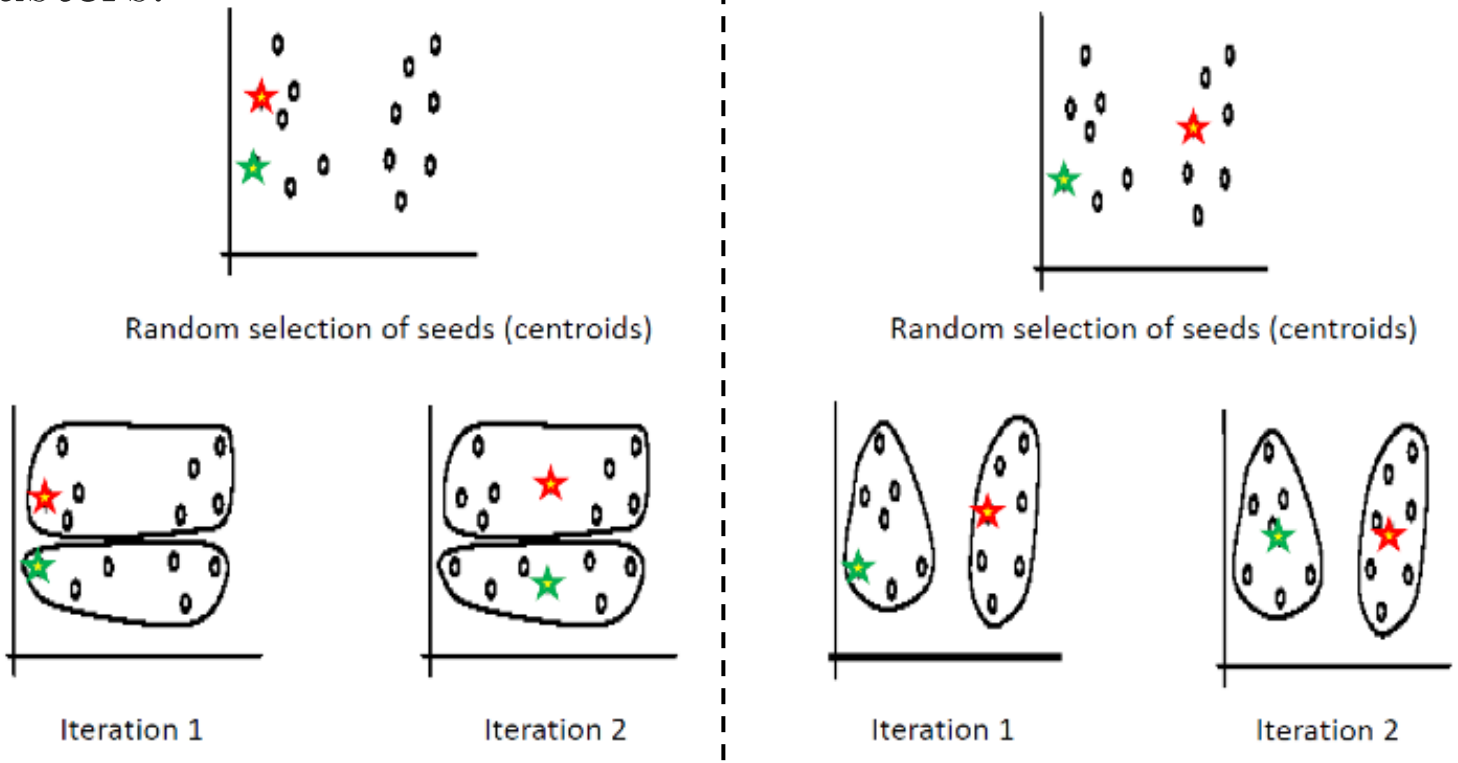


- Finding the global optimum is NP-hard.
- The *k*-means algorithm is guaranteed to converge to a local optimum.

Q: What causes the uncertainty?

# Sensitivity to Initial Centroids (Seeds)

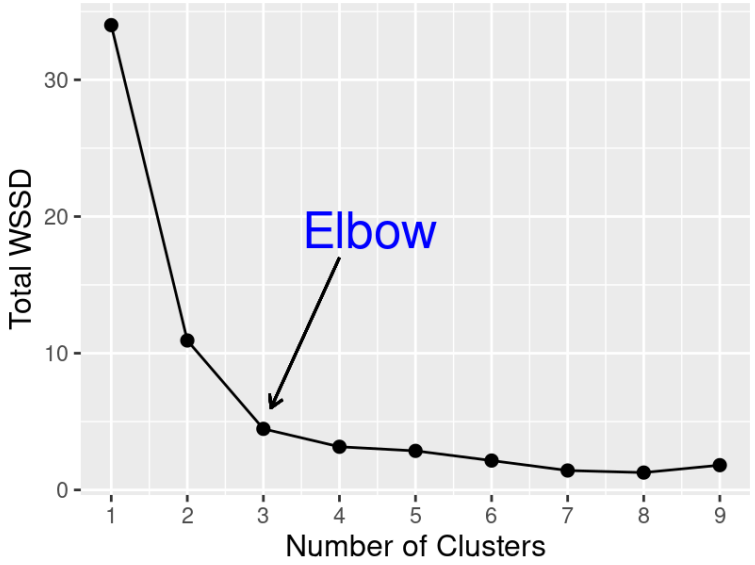
- Results can vary based on random seed selection.
- Some seeds can result in slow convergence or sub-optimal clusters.



# How to choose the value of K?

- The Elbow method is the best way to find K.
- For each K, the *within-sum-of-squares* (WSS) is defined as the sum of the squared distance between each member of the cluster and its centroid.
- K with the least amount of WSS is taken as the optimum.

$$WSS = \sum_{i=1}^{m} (x_i - c_i)^2$$



# How to choose the initial centroids?



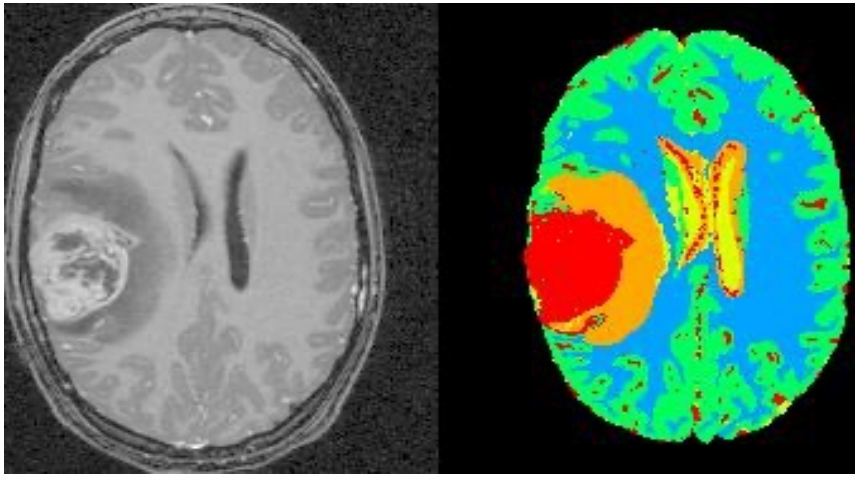

- Select good seeds using a heuristic or the results of another method.

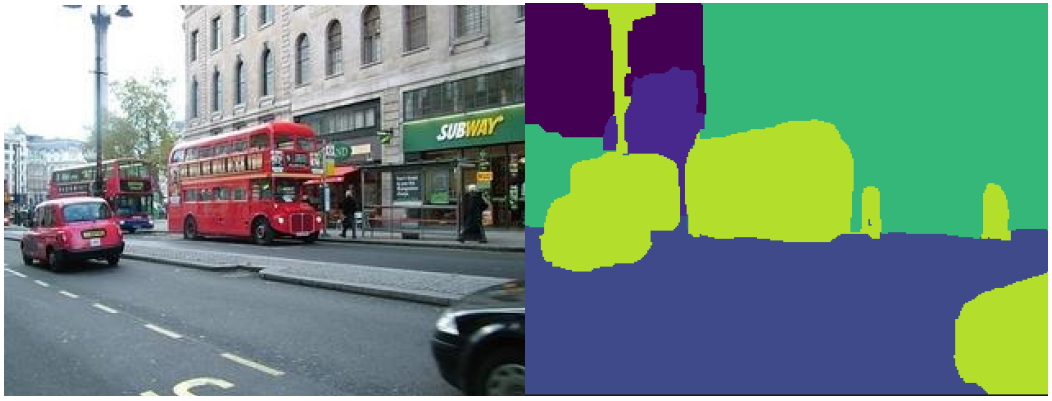
Example: k-means++

1. Choose one center uniformly at random from among the data points.
2. For each data point x, compute D(x), the distance between x and the nearest center that has already been chosen.
3. Choose one new data point at random as a new center, using a weighted probability distribution where a point x is chosen with probability proportional to D(x)^2 .
4. Repeat Steps 2 and 3 until k centers have been chosen.
5. Proceed using standard k-means clustering.

# Applications of K-Means

## Image Segmentation:





# Applications of K-Means

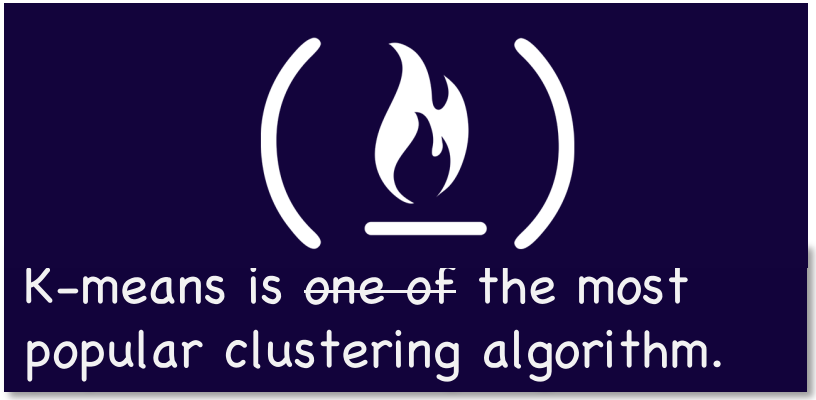
**Image Compression:** replace each pixel value with its nearby centroid



# Pros and Cons

Advantages:

- **Simple**: easy to understand and implement,
- **Efficient**: k-means is considered a linear algorithm.



# Pros and Cons

Disadvantages:

- Assume that each cluster is a sphere (球形)
- The algorithm is only applicable if the **mean** is defined.
- The user needs to specify $k$.
- The algorithm is sensitive to **outliers**.

What if we only know the **pairwise distances** of data?

# TIME for Some Coding

https://www.kaggle.com/code/adinishad/kmeans-clustering-from-scratch

https://www.kaggle.com/code/satishgunjal/tutorial-k-means-clustering

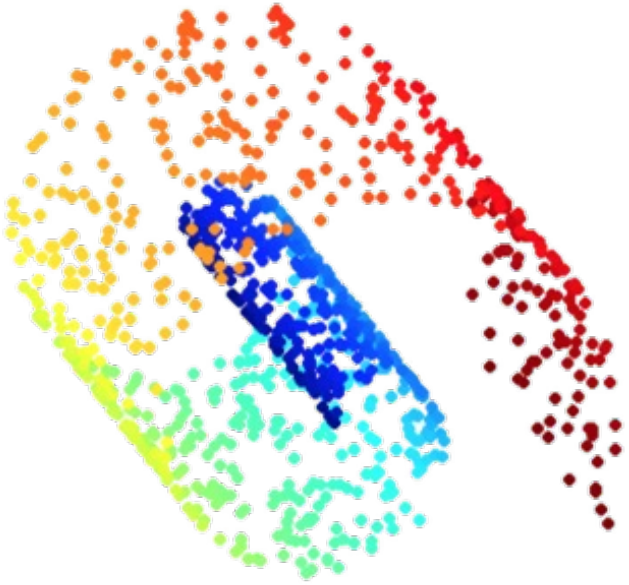


# What's Next?




## Dimensionality Reduction

Looks 3D



Actually 2D

